Updates

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I. Optimization Modifications Across the Three Deep2S DI-Attack Variants

For the Deep2S attack, I updated how the complex gains A_p are optimized in order to obtain a more stable attack under the same physical amplitude constraint $|A_p| \le A_{\max}$.

Old version (baseline). In the original implementation, I optimized the real and imaginary parts of the gains directly:

$$A_{\mathrm{re}} \leftarrow A_{\mathrm{re}} - \eta_{\mathrm{re}} \nabla_{A_{\mathrm{re}}} \mathcal{L}, \qquad A_{\mathrm{im}} \leftarrow A_{\mathrm{im}} - \eta_{\mathrm{im}} \nabla_{A_{\mathrm{im}}} \mathcal{L},$$

using relatively large learning rates. After each gradient step, I enforced the amplitude constraint via a hard projection: for each aperture p, I computed $|A_p|$ and, whenever $|A_p| > A_{\max}$, rescaled A_p back to $|A_p| = A_{\max}$. This procedure caused many entries to sit exactly at the bound $|A_p| = A_{\max}$ and led to oscillatory loss behavior due to the combination of large step sizes and abrupt clipping.

New version (current) In the revised implementation, I optimize unconstrained latent variables $Z_{\mathrm{re}}, Z_{\mathrm{im}}$ and map them smoothly to A:

$$A_{\mathrm{re}} \;=\; \frac{A_{\max}}{\sqrt{2}} \tanh\big(Z_{\mathrm{re}}\big), \qquad A_{\mathrm{im}} \;=\; \frac{A_{\max}}{\sqrt{2}} \tanh\big(Z_{\mathrm{im}}\big),$$

so that the complex gains are

$$A = A_{\mathrm{re}} + j A_{\mathrm{im}}.$$

This guarantees $|A_p| \le A_{\max}$ by construction (no explicit clipping) and preserves useful gradients near the boundary. Instead of hand-rolled gradient descent, I now use the Adam optimizer on $Z_{\mathrm{re}}, Z_{\mathrm{im}}$, which provides adaptive step sizes and smoother convergence. I retain an ℓ_2 penalty on $|A|$ (through the parameter `lambda_L2`) so that the optimizer does not simply push all gains to their maximum amplitude, but rather uses the available power budget in a more controlled and structured way.

In this revised formulation, the attack loss $\mathcal{L}$ is still computed in the image domain based on A (through the attacked measurements, Deep2S forward model, and MSE to the target volume), but during backpropagation we differentiate with respect to $Z_{\mathrm{re}}, Z_{\mathrm{im}}$. By the chain rule,

$$\frac{\partial \mathcal{L}}{\partial Z_{\mathrm{re}}} = \frac{\partial \mathcal{L}}{\partial A_{\mathrm{re}}} \cdot \frac{\partial A_{\mathrm{re}}}{\partial Z_{\mathrm{re}}} = \frac{\partial \mathcal{L}}{\partial A_{\mathrm{re}}} \frac{A_{\max}}{\sqrt{2}} \big(1 - \tanh^2(Z_{\mathrm{re}})\big),$$

$$\frac{\partial \mathcal{L}}{\partial Z_{\mathrm{im}}} = \frac{\partial \mathcal{L}}{\partial A_{\mathrm{im}}} \cdot \frac{\partial A_{\mathrm{im}}}{\partial Z_{\mathrm{im}}} = \frac{\partial \mathcal{L}}{\partial A_{\mathrm{im}}} \frac{A_{\max}}{\sqrt{2}} \big(1 - \tanh^2(Z_{\mathrm{im}})\big).$$

The optimizer (Adam in our implementation) then updates $Z_{\mathrm{re}}, Z_{\mathrm{im}}$:

$$Z_{\mathrm{re}} \leftarrow Z_{\mathrm{re}} - \alpha \,\mathrm{Adam}(\nabla_{Z_{\mathrm{re}}} \mathcal{L})$$

$$Z_{\mathrm{im}} \leftarrow Z_{\mathrm{im}} - \alpha \,\mathrm{Adam}(\nabla_{Z_{\mathrm{im}}} \mathcal{L})$$

Algorithm 1 Compressed Sensing Algorithm (CSA) in H, x notation

Require: Raw echo $\mathbf{y}_s$, measurement matrix $\mathbf{H}$
Ensure: 3-D imaging result cube $\hat{\mathbf{x}}$

Initialize parameters λ_0, p, and η, iterative threshold ε_0
$\hat{\mathbf{x}}^{(0)} = \mathbf{H}^{\mathrm{H}} \mathbf{y}_s$, set $n = 0$
$r = \frac{\|\hat{\mathbf{x}}^{(n)} - \hat{\mathbf{x}}^{(n-1)}\|_2}{\|\hat{\mathbf{x}}^{(n)}\|_2}$ ▷ defined for $n \ge 1$
while $r \ge \varepsilon_0$ **do**
  Diagonal matrix $\mathbf{\Lambda}^{(n)}$ with entries

$$\Lambda_{ii}^{(n)} = \frac{p}{2} \left(|\hat{x}_i^{(n-1)}|^2 + \eta \right)^{\frac{p}{2} - 1}$$

  Update image estimate

$$\hat{\mathbf{x}}^{(n)} = \left(\mathbf{H}^{\mathrm{H}} \mathbf{H} + \lambda_0 \, \beta^{(n)} \mathbf{\Lambda}^{(n)} \right)^{-1} \mathbf{H}^{\mathrm{H}} \mathbf{y}_s$$

  Update noise parameter

$$\beta^{(n)} = \frac{\|\mathbf{y}_s - \mathbf{H} \hat{\mathbf{x}}^{(n)}\|_2^2}{N}$$

  $n \leftarrow n + 1$
  $r = \frac{\|\hat{\mathbf{x}}^{(n)} - \hat{\mathbf{x}}^{(n-1)}\|_2}{\|\hat{\mathbf{x}}^{(n)}\|_2}$
end while
$\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}}^{(n)}$

or in general

$$Z_{\mathrm{re}}^{(n+1)} = Z_{\mathrm{re}}^{(n)} \;-\; \eta \, \frac{A_{\max}}{\sqrt{2}} \, \frac{\partial \mathcal{L}}{\partial A_{\mathrm{re}}} \Big(1 - \tanh^2\big(Z_{\mathrm{re}}^{(n)}\big)\Big),$$

$$Z_{\mathrm{im}}^{(n+1)} = Z_{\mathrm{im}}^{(n)} \;-\; \eta \, \frac{A_{\max}}{\sqrt{2}} \, \frac{\partial \mathcal{L}}{\partial A_{\mathrm{im}}} \Big(1 - \tanh^2\big(Z_{\mathrm{im}}^{(n)}\big)\Big),$$

and after each update we recompute $A_{\mathrm{re}}, A_{\mathrm{im}}$ via the $\tanh$ mapping. This guarantees $|A_p| \le A_{\max}$ by construction, while still allowing gradients to flow smoothly near the amplitude bound.

Overall, the change is from a "direct + hard-clipping" optimization of A with large steps to a smoothly constrained, Adam-based optimization in the latent space $(Z_{\mathrm{re}}, Z_{\mathrm{im}})$, under the same physical amplitude bound $|A_p| \le 2$.

II. "Linear Tikhonov Inverse" Surrogate In DI-Attack on CSA

In the CSA formulation, SAR measurements are modeled as

$$\mathbf{y}_s = \mathbf{H}\mathbf{x} + \mathbf{n}, \tag{1}$$

where $\mathbf{y}_s \in \mathbb{C}^{M_{\mathrm{meas}}}$ are the echoes, $\mathbf{x} \in \mathbb{C}^{N_{\mathrm{pix}}}$ is the unknown reflectivity, $\mathbf{H}$ is the propagation matrix, and $\mathbf{n}$ is noise.

Sparse prior and MAP objective CSA assumes a sparse scene,

$$p(\mathbf{x}) \propto \prod_{i=1}^{N_{\text{pix}}} \exp(-\lambda_0 |x_i|^p), \qquad 0 < p \le 1, \tag{2}$$

leading to the MAP problem

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \big\|\mathbf{y}_s - \mathbf{H}\mathbf{x}\big\|_2^2 + \lambda_0 \beta \|\mathbf{x}\|_p^p, \tag{3}$$

where β is an inverse noise–variance estimate.

SBRIM iterations (original CSA) CSA solves (3) via SBRIM. At iteration n,

$$\Lambda_{ii}^{(n)} = \frac{p}{2}\big(|x_i^{(n-1)}|^2 + \eta\big)^{\frac{p}{2}-1}, \tag{4}$$

$$x^{(n)} = \Big(\mathbf{H}^H\mathbf{H} + \lambda_0 \beta^{(n)} \Lambda^{(n)}\Big)^{-1} \mathbf{H}^H \mathbf{y}_s, \tag{5}$$

$$\beta^{(n)} = \frac{\|\mathbf{y}_s - \mathbf{H}x^{(n)}\|_2^2}{M_{\text{meas}}}. \tag{6}$$

Iterations stop when $r^{(n)} = \|x^{(n)} - x^{(n-1)}\|_2 / \|x^{(n)}\|_2 < \varepsilon_0$, and the final CSA reconstruction is $\hat{\mathbf{x}} = x^{(n_{\text{final}})}$. Each step solves a regularized least-squares system with an adaptively reweighted diagonal penalty. Large coefficients are penalized less and small ones more, promoting sparsity in $\hat{\mathbf{x}}$.

Original CSA vs. Our Attack Implementation

Victim mapping: the victim we attack is the original CSA/SBRIM mapping

$$\mathbf{y}_s \;\mapsto\; \hat{\mathbf{x}}_{\text{CSA}} = \text{SBRIM}\big(\mathbf{y}_s, \mathbf{H}, \lambda_0, p, \eta\big),$$

implemented exactly as in the equations above.

We inject a complex gain vector $A \in \mathbb{C}^{N_p}$ into a bank of time–domain attack waveforms $D \in \mathbb{C}^{N_{\text{samp}} \times N_p}$, added to the clean raw data $X_v \in \mathbb{C}^{N_{\text{samp}} \times N_p}$. After SAR preprocessing (FFT, range-bin selection, row flipping), this yields attacked CSA measurements

$$\mathbf{y}_s^{\text{atk}}(A) = G_{pre}(X_v + D\,\text{diag}(A)), \tag{7}$$

where operator G_{pre} represent all pre-processing. Now, our objective is to optimized A so that the CSA output matches a desired target image x_{tar}.

Differentiable surrogate for optimization. Backpropagating through SBRIM is expensive and slow, so during optimization we replace CSA by a linear Tikhonov inverse operator built from the same $\mathbf{H}$. Starting from the regularized least-squares problem

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \big\|\mathbf{y}_s^{\text{atk}} - \mathbf{H}\mathbf{x}\big\|_2^2 + \lambda_{\text{lin}} \|\mathbf{x}\|_2^2, \tag{8}$$

the normal equations are

$$\big(\mathbf{H}^H\mathbf{H} + \lambda_{\text{lin}} I\big)\hat{\mathbf{x}} = \mathbf{H}^H \mathbf{y}_s^{\text{atk}}. \tag{9}$$

Solving for $\hat{\mathbf{x}}$ yields the "linear Tikhonov inverse"

$$W_{\text{CSA}} = \big(\mathbf{H}^H\mathbf{H} + \lambda_{\text{lin}} I\big)^{-1} \mathbf{H}^H, \tag{10}$$

so that, for any attacked measurement vector $\mathbf{y}_s^{\text{atk}}(A)$, the surrogate image is

$$\tilde{x}(A) = W_{\text{CSA}}\, \mathbf{y}_s^{\text{atk}}(A). \tag{11}$$

We then optimize the complex gains A via

$$A^\star = \arg\min_A \Big(\|\tilde{x}(A) - x_{\text{tar}}\|_2^2 + \lambda_{\text{L2}} \|A\|_2^2 \Big), \tag{12}$$

using gradient descent, where gradients $\partial\tilde{x}(A)/\partial A$ are obtained through the linear surrogate W_{CSA}.

Final evaluation (true CSA attack). After optimization, we form

$$\mathbf{y}_s^{\text{atk}} = \mathbf{y}_s^{\text{atk}}(A^\star) \tag{13}$$

and feed it into the *original* CSA/SBRIM algorithm:

$$\hat{x}_{\text{CSA}}^{\text{atk}} = \text{SBRIM}\big(\mathbf{y}_s^{\text{atk}}, \mathbf{H}, \lambda_0, p, \eta\big). \tag{14}$$

Thus, W_{CSA} is used only as a gradient surrogate, whereas the attack performance is always measured on the full CSA reconstructor.

III. BPA Surrogate in DI-Attack on LIA

Original LIA (victim mapping). For near-field SAR, let $\mathbf{y}_s \in \mathbb{C}^{M_{\text{meas}}}$ be the vectorized measurements and $\mathbf{x} \in \mathbb{C}^{N_{\text{pix}}}$ the unknown reflectivity image. As in BPA, we build a propagation matrix $\mathbf{H}_{\text{BPA}} \in \mathbb{C}^{M_{\text{meas}} \times N_{\text{pix}}}$ from the sensor and pixel geometry. LIA (Li & Chen) then uses a subsampled version

$$\mathbf{H}_p = \mathbf{H}_{\text{BPA}} \in \mathbb{C}^{K \times N_{\text{pix}}},$$

where $\mathcal{P}$ indexes a subset of K measurements, and applies an iterative matrix–inversion–lemma update to reconstruct

$$\hat{x}_{\text{LIA}} = \text{LIA}\big(\mathbf{y}_s, \mathbf{H}_p\big).$$

As in the CSA case, we inject a complex gain vector $A \in \mathbb{C}^{N_p}$ on top of the clean raw data $X_v \in \mathbb{C}^{N_{\text{samp}} \times N_p}$ via a bank of time–domain attack waveforms $D \in \mathbb{C}^{N_{\text{samp}} \times N_p}$:

$$\mathbf{y}_s^{\text{atk}}(A) = G_{pre}\big(X_v + D\,\text{diag}(A)\big), \tag{15}$$

The victim we are attacking is the original LIA reconstructor

$$\mathbf{y}_s \;\mapsto\; \hat{\mathbf{x}}_{\text{LIA}} = \text{LIA}\big(\mathbf{y}_s, \mathbf{H}_p\big).$$

BPA surrogate for gradients. Backpropagating through the full LIA iterations is costly, so during the optimization we replace LIA with a simple BPA-style linear inverse built from the same propagation matrix $W_{\text{BPA}} = \mathbf{H}_{\text{BPA}}^H$, i.e., standard back-projection. For any candidate A, the surrogate image is

$$\tilde{\mathbf{x}}(A) = W_{\text{BPA}}\, \mathbf{y}_s^{\text{atk}}(A), \tag{16}$$

and we solve

$$A^\star = \arg\min_A \Big(\|\tilde{\mathbf{x}}(A) - \mathbf{x}_{\text{tar}}\|_2^2 + \lambda_{\text{L2}} \|A\|_2^2 \Big) \tag{17}$$

by gradient descent, using W_{BPA} as a differentiable surrogate for LIA$(\cdot)$. After convergence, we form the attacked measurements

$$\mathbf{y}_s^{\text{atk}} = \mathbf{y}_s^{\text{atk}}(A^\star) \tag{18}$$

and feed them into the original LIA algorithm:

$$\hat{\mathbf{x}}_{\text{LIA}}^{\text{atk}} = \text{LIA}\big(\mathbf{y}_s^{\text{atk}}, \mathbf{H}_p\big). \tag{19}$$

Thus, as with CSA, BPA is used *only* as a surrogate for gradients, while the actual victim is the full iterative LIA reconstructor.